

Pharma Vision 360

End-to-End Pharmaceutical
Analytics Using Descriptive,
Diagnostic, Predictive &
Prescriptive Intelligence

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Project Objectives

Analyze pharmaceutical sales performance

Monitor inventory efficiency

Detect expiry risks

Evaluate supplier performance

Forecast future medicine demand

Build operational dashboards

Business Problem Statement

Pharmaceutical organizations frequently face operational challenges such as fluctuating medicine demand, inventory shortages, supplier delays, expiry-related losses, and inconsistent revenue performance. The absence of integrated analytical monitoring often leads to inefficient inventory management, disrupted supply chains, and reduced operational efficiency.

This project aims to develop an end-to-end Pharmaceutical Analytics Framework using Python, SQL, and Power BI to analyze sales performance, monitor inventory health, evaluate supplier efficiency, identify operational risks, and forecast future demand trends.

By leveraging Descriptive, Diagnostic, Predictive, and Prescriptive Analytics, the solution transforms raw pharmaceutical data into actionable business intelligence, enabling data-driven decision-making, improved supply chain visibility, optimized inventory management, and enhanced operational performance.

Tools & Technologies Used

Tool	Purpose
Python	Data Analysis
Pandas	Data Cleaning
NumPy	Numerical Operations
Matplotlib	Visualization
Seaborn	Statistical Analysis
Statsmodels	Forecasting
MySQL	Database Analytics
Power BI	Dashboarding
Jupyter Notebook	Development

Datasets

Tables:

- Sales
- Medicines
- Inventory
- Warehouses
- Suppliers

Data Cleaning & Preparation

- missing values
- datatype conversion
- date formatting
- merging tables
- feature engineering

Exploratory Data Analysis (EDA)

1. Monthly Revenue Trend Analysis

Key Insight

Revenue reached its highest point at approximately: **325K+** during Month 3, followed by a sharp decline to nearly: **100K** by 7 Month

The later months demonstrate: **stable recovery and operational normalization**

The monthly revenue trend shows:

- strong revenue growth during the initial months
- peak performance in Month 3
- a significant revenue decline between Months 4–7
- gradual business recovery from Month 8 onward

Possible Causes

Early Revenue Growth (Months 1–3)

Possible drivers:

- high medicine demand
- seasonal healthcare requirements
- successful supplier availability
- strong inventory support
- effective warehouse operations

Revenue Decline (Months 4–7)

The sharp drop may indicate:

- inventory shortages
- supplier delays
- medicine stockouts
- seasonal reduction in pharmaceutical demand
- operational disruptions within supply chain

The decline pattern suggests:

possible operational inefficiency rather than random fluctuation

Recovery Phase (Months 8–12)

The gradual upward trend indicates:

- improved stock replenishment
- stabilized supplier performance
- operational recovery
- increased medicine demand consistency

However, revenue did not fully recover to the Month 3 peak, suggesting:

partial recovery with remaining optimization opportunities

Business Impact

Negative Impacts During Decline Phase

The revenue drop likely resulted in:

- reduced profitability
- inefficient warehouse utilization
- lower operational efficiency
- potential customer demand loss
- supply chain instability

Positive Recovery Indicators

The recovery trend indicates:

- improved operational management
- better inventory availability
- stabilization of pharmaceutical sales cycles

Predictive Analysis / Future Outlook

Based on the observed trend:

- revenue is expected to continue gradual stabilization
- future growth is likely if inventory continuity is maintained
- supplier optimization may accelerate revenue recovery
- seasonal fluctuations may continue affecting performance

If current recovery patterns continue:

future revenue may exceed previous peak levels

particularly with improved demand forecasting and inventory planning.

Recommendations

Inventory Optimization

Implement proactive inventory monitoring to prevent stock shortages during high-demand periods.

Supplier Performance Management

Reduce supply-chain disruptions by:

- monitoring supplier delays
- improving lead-time management
- diversifying supplier dependency

Seasonal Planning

Prepare operational strategies for expected seasonal fluctuations by:

- increasing stock before peak demand periods
- optimizing warehouse allocation
- improving replenishment cycles

Revenue Recovery Strategy

Focus on:

- high-performing medicine categories
- warehouse optimization
- operational continuity

to accelerate sustained revenue growth.

2.Trend & Seasonality Analysis

Key Insight

The decomposition analysis highlights:

- a declining revenue trend during the mid-period,
- gradual recovery in later months,
- and recurring seasonal demand fluctuations across the year.

The trend component indicates long-term business movement, while the seasonal component confirms:

repetitive pharmaceutical demand patterns during specific periods.

The residual component shows irregular fluctuations caused by operational or external business factors.

Possible Causes

Revenue Trend Decline

The mid-period decline may be associated with:

- inventory shortages,
- supplier delays,
- reduced operational efficiency,
- or lower seasonal medicine demand.

Seasonal Variations

The recurring seasonal peaks and drops suggest:

- periodic medicine demand cycles,
- healthcare seasonality,
- and recurring purchasing behavior.

This indicates:

predictable business seasonality within pharmaceutical operations.

Business Impact

Operational Impact

Revenue instability may lead to:

- inefficient inventory utilization,
- warehouse imbalance,
- supply chain disruptions,
- and reduced operational efficiency.

Financial Impact

Seasonal fluctuations directly affect:

- revenue consistency,
- profit stability,
- and resource allocation planning.

Understanding seasonality helps businesses:

optimize operational and financial planning more effectively.

Predictive Analysis / Future Outlook

The recovery trend suggests:

- gradual revenue stabilization,
- improved operational continuity,
- and potential future growth.

The seasonal component indicates that:

future demand fluctuations are likely to repeat

in similar business cycles.

With improved forecasting and inventory planning, future operational performance can become more stable and predictable.

Recommendations

Demand Forecasting

Implement predictive forecasting models to prepare for recurring seasonal demand changes.

Inventory Optimization

Increase inventory availability before high-demand seasonal periods to avoid stock shortages.

Supplier Performance Monitoring

Improve supplier coordination and lead-time management to reduce operational disruptions.

Seasonal Business Planning

Develop proactive operational strategies aligned with recurring pharmaceutical demand patterns.

Continuous Monitoring

Utilize real-time dashboards and statistical monitoring to track trend movement and seasonal performance continuously.

3. Seasonality Analysis

Key Insight

The seasonality analysis reveals relatively stable pharmaceutical revenue patterns across most months, with peak average revenue observed during Months 2–4 and comparatively lower revenue during Months 8–9.

The analysis indicates:

consistent cyclical demand behavior with moderate seasonal fluctuations rather than extreme volatility.

Possible Causes

Higher Seasonal Revenue

The higher revenue periods may be influenced by:

- increased medicine demand,
- seasonal healthcare requirements,
- higher prescription volumes,
- or improved inventory availability.

Lower Seasonal Revenue

The decline during Months 8–9 may result from:

- reduced healthcare demand,
- inventory limitations,
- supplier disruptions,
- or operational slowdowns.

The recurring pattern suggests:

predictable pharmaceutical demand cycles throughout the year.

Business Impact

Operational Impact

Seasonal demand fluctuations directly influence:

- inventory planning,
- warehouse allocation,
- supplier coordination,
- and stock replenishment cycles.

Failure to manage seasonality effectively may lead to:

- stock shortages,
- overstocking,
- and inefficient resource utilization.

Financial Impact

Revenue variations affect:

- profitability consistency,
- operational budgeting,
- and supply chain efficiency.

Understanding seasonal demand helps businesses:

improve operational preparedness and
financial stability

during peak and low-demand periods.

Predictive Analysis / Future Outlook

The recurring seasonal pattern suggests that similar revenue fluctuations are likely to continue in future business cycles.

The relatively stable seasonal variation indicates:

- predictable demand behavior,
- improved forecasting opportunities,
- and better inventory optimization potential.

With proactive planning:

future seasonal disruptions can be
minimized

while maintaining revenue stability.

Recommendations

Seasonal Inventory Planning

Increase stock availability before peak-demand periods to prevent stock shortages.

Demand Forecasting

Utilize predictive analytics to forecast seasonal demand fluctuations more accurately.

Supplier Coordination

Strengthen supplier planning during high-demand months to ensure inventory continuity.

Warehouse Optimization

Allocate inventory strategically across warehouses based on expected seasonal demand.

Continuous Monitoring

Implement dashboard-based monitoring to track seasonal performance and operational

4.Revenue Trend with Moving Average Analysis

Key Insight

The analysis compares actual revenue performance with the 3-month moving average trend, highlighting a sharp revenue decline during the mid-period followed by gradual recovery in later months.

The moving average line smooths short-term fluctuations and reveals:

the underlying long-term revenue trend more clearly.

The actual revenue remained highly volatile during the middle months, while the moving average confirms a steady recovery trend toward the end of the year.

Possible Causes

Revenue Decline

The sharp decline in actual revenue may be caused by:

- inventory shortages,
- supplier delays,
- operational disruptions,
- or reduced pharmaceutical demand.

Revenue Recovery

The gradual upward movement after Month 7 suggests:

- improved inventory replenishment,
- stabilized supplier performance,
- and operational recovery.

The moving average indicates:

business stabilization after a volatile period

Business Impact

Operational Impact

Revenue volatility can negatively affect:

- inventory planning,
- warehouse efficiency,
- supplier coordination,
- and resource allocation.

Operational instability may also increase:

- stock imbalance,
- supply chain pressure,
- and forecasting inaccuracies.

Financial Impact

Fluctuating revenue patterns may lead to:

- inconsistent profitability,
- reduced financial predictability,
- and inefficient operational budgeting.

The moving average helps businesses:

identify long-term performance direction
beyond short-term fluctuations

Predictive Analysis / Future Outlook

The upward moving average trend indicates:

- improving revenue stability,
- gradual operational recovery,
- and positive future business outlook.

If current operational improvements continue:

future revenue is expected to maintain
steady growth

with reduced volatility over time.

The moving average also suggests stronger forecasting
reliability in upcoming business periods.

Recommendations

Forecast-Based Planning

Utilize moving average and forecasting techniques for proactive revenue and inventory planning.

Inventory Optimization

Maintain adequate stock levels during demand recovery periods to support revenue continuity.

Supplier Performance Monitoring

Improve supplier coordination to minimize operational disruptions affecting revenue stability.

Continuous Trend Monitoring

Track moving average trends regularly to identify early operational or revenue fluctuations.

Strategic Decision-Making

Use smoothed trend analysis to support long-term business planning and financial forecasting.

5.Monthly Revenue Growth Rate Analysis

Key Insight

The monthly revenue growth analysis shows significant fluctuations in pharmaceutical revenue performance throughout the year.

The business experienced:

- strong positive growth during Month 3,
- severe negative growth between Months 4–5,
- and a major recovery spike during Month 8.

The highest growth was observed in Month 8 with approximately:

80%+ revenue growth

indicating strong operational recovery after a prolonged decline period.

Possible Causes

Negative Growth Phase

The substantial revenue decline during Months 4–5 may be caused by:

- inventory shortages,
- supplier delays,
- operational disruptions,
- reduced seasonal demand,
- or inefficient stock management.

Recovery Phase

The sharp positive growth during Month 8 suggests:

- improved inventory replenishment,
- stabilization of supply chain operations,
- increased pharmaceutical demand,
- and operational recovery initiatives.

The fluctuating growth pattern indicates:

high revenue volatility within business operations

Business Impact

Operational Impact

Revenue instability may lead to:

- inefficient inventory planning,
- warehouse imbalance,
- forecasting challenges,
- and inconsistent supply chain performance.

Large fluctuations also increase:

- operational uncertainty,
- resource allocation difficulty,
- and demand planning risk.

Financial Impact

Negative growth periods directly affect:

- profitability,
- business stability,
- and financial forecasting accuracy.

The sharp recovery demonstrates:

strong business recovery capability
but also highlights dependence on operational
consistency.

Predictive Analysis / Future Outlook

The recovery trend after Month 8 suggests:

- improving business stability,
- positive future revenue outlook,
- and stronger operational continuity.

However, recurring fluctuations indicate that:

future volatility may continue if
operational risks are not controlled

With improved forecasting and supply chain optimization,
revenue growth can become more stable and
sustainable.

Recommendations

Revenue Forecasting

Implement predictive forecasting models to anticipate revenue fluctuations and improve planning accuracy.

Inventory Optimization

Strengthen inventory monitoring to reduce stock shortages during high-demand periods.

Supplier Performance Management

Improve supplier coordination and lead-time monitoring to minimize operational disruptions.

Risk Monitoring

Establish real-time operational monitoring systems to detect sudden revenue changes early.

Strategic Growth Planning

Develop proactive operational strategies focused on sustaining long-term revenue stability and controlled business growth.

6.Top 10 Medicines by Revenue Analysis

Key Insight

The analysis identifies the highest revenue-generating medicines within the pharmaceutical portfolio.

Among all medicines, **Calcium Tablet** generated exceptionally higher revenue compared to other products, significantly outperforming the remaining medicines.

The top-performing medicines include:

- Calcium Tablet
- Dolo 650
- Paracetamol
- Omeprazole
- Amoxicillin

The chart indicates:

strong revenue concentration among a limited number of medicines

with Calcium Tablet contributing a major share of total pharmaceutical revenue.

Possible Causes

High Revenue Medicines

The strong performance of top medicines may be influenced by:

- high market demand,
- frequent prescriptions,
- wider healthcare usage,
- seasonal medical requirements,
- and consistent inventory availability.

Revenue Gap Between Medicines

The substantial gap between Calcium Tablet and other medicines suggests:

- dominant product dependency,
- stronger customer demand concentration,
- or higher sales volume compared to other products.

This indicates:

uneven revenue distribution across medicine categories

Business Impact

Positive Business Impact

High-performing medicines contribute to:

- stronger revenue generation,
- increased profitability,
- improved sales stability,
- and better operational performance.

Operational Risk

Heavy dependence on a small group of medicines may create:

- revenue concentration risk,
- inventory pressure on high-demand medicines,
- and operational vulnerability if stock shortages occur.

This dependency may impact:

business continuity during supply chain disruptions

Predictive Analysis / Future Outlook

The current trend suggests that:

- high-demand medicines will likely continue driving future revenue,
- demand concentration may remain consistent,
- and top-performing medicines will require stronger inventory support.

If current patterns continue:

Calcium Tablet is expected to remain the primary revenue driver

while other medicines may require strategic promotion to diversify revenue streams.

Recommendations

Inventory Prioritization

Maintain higher stock availability for top-performing medicines to prevent revenue loss from stockouts.

Revenue Diversification

Promote mid-performing medicines to reduce dependency on a limited number of products.

Demand Forecasting

Use predictive analytics to forecast demand for high-revenue medicines and optimize inventory planning.

Supplier Coordination

Ensure stable supplier relationships for critical high-demand medicines to maintain uninterrupted availability.

Product Portfolio Optimization

Analyze low-performing medicines to identify opportunities for:

- pricing optimization,
- product repositioning,
- or inventory reduction.

7.Warehouse Revenue Performance Analysis

Key Insight

The warehouse revenue analysis reveals significant variation in revenue contribution across different warehouses.

Warehouses:

- WH007

- WH014

- WH015

generated the highest revenue, while warehouses such as:

- WH001

- WH002

contributed comparatively lower revenue.

The analysis indicates:

uneven operational and sales performance
across warehouse locations

with a small group of warehouses driving a major share of
total revenue.

Possible Causes

High-Performing Warehouses

The strong performance of leading warehouses may be due to:

- higher medicine demand in specific regions,
- better inventory availability,
- efficient warehouse operations,
- stronger supplier coordination,
- or strategic geographic location advantages.

Low-Performing Warehouses

Lower-performing warehouses may experience:

- reduced regional demand,
- inventory shortages,
- inefficient stock allocation,
- supplier delays,
- or operational inefficiencies.

This performance gap suggests:

imbalanced warehouse utilization across
the supply chain

Business Impact

Positive Impact

High-performing warehouses contribute significantly to:

- revenue generation,
- operational continuity,
- faster inventory movement,
- and supply chain efficiency.

Operational Risk

Heavy dependency on a limited number of warehouses may create:

- operational bottlenecks,
- inventory pressure,
- increased supply chain vulnerability,
- and regional distribution imbalance.

Underperforming warehouses may also lead to:

inefficient resource utilization and
reduced operational efficiency

Predictive Analysis / Future Outlook

The current trend suggests that:

- top-performing warehouses are likely to continue driving future revenue,
- operational imbalance may persist if warehouse optimization is not implemented,
- and low-performing warehouses may continue underutilization.

If operational improvements are introduced:

overall warehouse productivity and revenue distribution can become more balanced across the pharmaceutical network.

Recommendations

Warehouse Optimization

Improve inventory allocation and operational planning for low-performing warehouses.

Demand-Based Inventory Distribution

Allocate stock strategically based on regional demand patterns and warehouse performance.

Operational Performance Monitoring

Continuously track warehouse KPIs such as:

- revenue contribution,
- inventory turnover,
- and stock availability.

Supplier & Logistics Coordination

Enhance logistics efficiency and supplier coordination for underperforming warehouse locations.

Risk Diversification

Reduce dependency on a few high-performing warehouses by strengthening operational capacity across all warehouse locations.

8.Expiry Risk Distribution Analysis

Key Insight

The expiry risk analysis indicates that the majority of pharmaceutical inventory is currently categorized as **Safe**, accounting for approximately:

96.7% of total inventory

while only a small portion:

3.3%

falls under the **High-Risk** expiry category.

This suggests:

strong overall inventory stability and effective expiry management across pharmaceutical operations.

Possible Causes

High Safe Inventory Percentage

The high proportion of safe inventory may result from:

- effective inventory monitoring,
- proper stock rotation,
- efficient warehouse management,
- and timely inventory replenishment.

Presence of High-Risk Inventory

The small high-risk segment may be caused by:

- slow-moving medicines,
- excess stock accumulation,
- uneven warehouse distribution,
- or reduced demand for certain medicines.

This indicates:

potential inventory holding inefficiencies in specific product segments

Business Impact

Positive Operational Impact

A low expiry risk percentage contributes to:

- reduced inventory wastage,
- improved stock utilization,
- higher operational efficiency,
- and better financial performance.

Potential Risk Impact

Even a small percentage of high-risk inventory may lead to:

- financial losses from expired medicines,
- increased storage costs,
- and reduced inventory turnover efficiency.

If unmanaged:

expiry-related losses may gradually
increase over time

particularly for low-demand medicines.

Predictive Analysis / Future Outlook

The current inventory trend suggests:

- stable inventory health,
- effective warehouse operations,
- and controlled expiry risk levels.

However, recurring overstocking or slow-moving inventory may increase future expiry risk if continuous monitoring is not maintained.

With proactive inventory optimization:

future expiry-related losses can remain
minimal

while maintaining operational efficiency.

Recommendations

Inventory Monitoring

Implement continuous inventory tracking to identify high-risk medicines early.

FEFO Inventory Strategy

Adopt:

First Expiry First Out (FEFO)

inventory management practices to minimize expiry losses.

Demand-Based Stock Planning

Optimize stock allocation based on medicine demand patterns to reduce overstocking.

Slow-Moving Medicine Analysis

Regularly identify and monitor slow-moving medicines with higher expiry risk exposure.

Automated Expiry Alerts

Implement automated dashboard alerts for medicines approaching expiry thresholds.

9. Stock Status Analysis

Key Insight

The stock status analysis shows that all medicines in the inventory are currently categorized under:

"Sufficient Stock"

indicating stable inventory availability across the pharmaceutical supply chain.

No medicines were identified under low-stock or critical-stock categories, suggesting:

effective inventory management and stock planning

within warehouse operations.

Possible Causes

The strong stock availability may be driven by:

- efficient inventory replenishment,
- accurate stock forecasting,
- proper warehouse monitoring,
- and stable supplier performance.

The absence of low-stock medicines indicates:

well-maintained operational continuity across inventory systems.

Business Impact

Positive Operational Impact

Sufficient inventory levels contribute to:

- uninterrupted medicine availability,
- improved customer satisfaction,
- reduced stockout risk,
- and smoother warehouse operations.

Financial Impact

Stable inventory management helps:

- maintain revenue continuity,
- reduce emergency procurement costs,
- and improve operational efficiency.

However, consistently high stock levels may also create:

potential overstocking and inventory
holding risks

if demand forecasting is not optimized.

Predictive Analysis / Future Outlook

The current inventory trend suggests:

- stable operational performance,
- strong supply chain continuity,
- and reduced short-term stock shortage risk.

If inventory growth continues without proportional demand:

future overstocking risk may increase
particularly for slow-moving medicines.

With balanced inventory optimization, future stock stability
can remain efficient and sustainable.

Recommendations

Inventory Optimization

Continuously monitor inventory levels to maintain balance between stock availability and demand.

Demand Forecasting

Use predictive analytics to align stock replenishment with expected pharmaceutical demand.

Warehouse Monitoring

Track inventory movement regularly to identify slow-moving or excess stock early.

Supplier Coordination

Maintain stable supplier relationships to ensure efficient replenishment without overstocking.

Automated Inventory Alerts

Implement real-time monitoring systems for proactive stock management and operational control.

10. Supplier Lead Time Analysis

Key Insight

The supplier lead time analysis reveals noticeable variation in delivery timelines across pharmaceutical suppliers.

Suppliers such as:

- **BioHealth Agencies**
- **Lifeline Suppliers**
- **Medicore Traders**

show the highest lead times, while suppliers like:

- **Medisupply Ltd**
- **Zenith Medicos**
- **VitalCare Pharma**

maintain comparatively lower delivery durations.

The analysis indicates:

uneven supplier delivery performance across the supply chain

which may directly influence inventory availability and operational continuity.

Possible Causes

High Lead Time Suppliers

Longer lead times may result from:

- transportation delays,
- inefficient logistics,
- higher supplier workload,
- geographic distance,
- or operational bottlenecks.

Low Lead Time Suppliers

Suppliers with shorter lead times likely benefit from:

- efficient logistics management,
- stable operational processes,
- stronger distribution networks,
- and better delivery coordination.

This variation suggests:

inconsistent supplier operational efficiency within the pharmaceutical procurement system.

Business Impact

Higher supplier lead times may lead to:

- delayed inventory replenishment,
- stock shortages,
- warehouse disruptions,
- and operational inefficiencies.

Financial Impact

Supplier delays can negatively affect:

- revenue continuity,
- inventory planning accuracy,
- customer satisfaction,
- and supply chain reliability.

Dependence on slow-performing suppliers may increase: overall operational and procurement risk across pharmaceutical operations.

Predictive Analysis / Future Outlook

The current trend suggests that:

- suppliers with consistently high lead times may continue creating operational delays,
- inventory pressure may increase during high-demand periods,
- and supply chain efficiency may remain uneven without supplier optimization.

If supplier performance monitoring improves:

future delivery consistency and inventory stability can significantly improve across the pharmaceutical network.

Recommendations

Supplier Performance Monitoring

Continuously evaluate supplier lead times to identify operational inefficiencies early.

Supplier Diversification

Reduce dependency on high lead-time suppliers by strengthening alternative supplier networks.

Procurement Optimization

Implement proactive procurement planning based on supplier delivery performance.

Inventory Buffer Strategy

Maintain safety stock for medicines sourced from suppliers with longer lead times.

Logistics Coordination

Improve transportation and logistics management to reduce supplier delivery delays and improve operational continuity.

11. Correlation Analysis Between Quantity Sold & Revenue

Key Insight

The correlation matrix shows a very strong positive correlation of:

0.92

between:

- **Quantity Sold**

- **and Revenue**

This indicates that revenue growth is highly dependent on the volume of medicines sold.

The analysis confirms:

strong direct relationship between sales volume and business revenue within pharmaceutical operations.

Possible Causes

The strong positive correlation may be driven by:

- high demand for pharmaceutical products,
- consistent medicine pricing,
- stable sales performance,
- and effective inventory availability.

As medicine sales increase:

revenue increases proportionally showing operational consistency across sales activities.

Business Impact

Operational Impact

The strong relationship between sales quantity and revenue helps businesses:

- forecast revenue more accurately,
- optimize inventory planning,
- and improve demand management.

Financial Impact

Higher medicine sales directly contribute to:

- stronger revenue generation,
- improved profitability,
- and better financial stability.

However, heavy dependency on sales volume may also create:

revenue vulnerability during demand
fluctuations
or operational disruptions.

Predictive Analysis / Future Outlook

The strong correlation suggests that:

- future revenue trends can be effectively predicted using sales volume data,
- demand forecasting models may achieve high accuracy,
- and inventory planning can become more data-driven.

If sales growth continues:

future revenue is expected to increase
consistently
provided inventory and supply chain continuity are
maintained.

Recommendations

Demand Forecasting

Use sales quantity trends for predictive revenue forecasting and operational planning.

Inventory Optimization

Maintain adequate inventory levels to support high-demand medicine sales.

Sales Performance Monitoring

Track medicine sales continuously to identify revenue growth opportunities early.

Revenue Diversification

Reduce dependency on volume-based revenue by optimizing:

- pricing strategies,
- product mix,
- and category performance.

Data-Driven Decision Making

Leverage correlation insights for strategic planning across:

- inventory management,
- warehouse operations,
- and revenue optimization.

12.Sales Anomaly Detection Analysis

Key Insight

The anomaly detection analysis identifies unusual spikes in medicine sales volume compared to normal sales patterns. Most sales records remain within a stable operational range, while a small number of observations were detected as: sales anomalies or outliers with exceptionally high quantity sold values. These anomalies indicate sudden surges in pharmaceutical demand beyond normal business behavior.

Possible Causes

The abnormal sales spikes may be influenced by:

- sudden increases in medicine demand,
- seasonal healthcare emergencies,
- bulk pharmaceutical purchases,
- promotional activities,
- or regional healthcare requirements.

Certain anomalies may also result from:

- inventory reporting inconsistencies,
- operational irregularities,
- or unexpected supply chain events.

The pattern suggests:

non-standard sales behavior within specific operational periods

Business Impact

Positive Business Impact

High sales spikes may contribute to:

- increased short-term revenue,
- improved inventory turnover,
- and stronger market demand performance.

Operational Risk

Unexpected sales anomalies may also create:

- inventory shortages,
- warehouse pressure,
- inaccurate demand forecasting,
- and supply chain disruptions.

If unmanaged:

sudden demand surges may negatively impact operational stability particularly during high-demand periods.

Predictive Analysis / Future Outlook

The recurring presence of anomalies suggests that:

- future unexpected demand spikes may continue,
- operational volatility may increase during peak demand cycles,
- and inventory pressure may rise without proactive forecasting.

With anomaly-based monitoring systems:

future demand fluctuations can be identified earlier and managed proactively improving operational preparedness.

Recommendations

Real-Time Sales Monitoring

Implement continuous sales monitoring systems to detect unusual demand spikes immediately.

Inventory Buffer Planning

Maintain safety stock for high-demand medicines to handle sudden sales surges effectively.

Predictive Demand Forecasting

Use anomaly-aware forecasting models to improve demand prediction accuracy.

Operational Risk Management

Investigate abnormal sales behavior regularly to identify:

- demand surges,
- operational issues,
- or reporting inconsistencies.

Supply Chain Preparedness

Strengthen supplier coordination and warehouse responsiveness during abnormal demand periods to maintain operational continuity.

13.Demand Forecast Analysis

Key Insight

The demand forecasting analysis compares historical pharmaceutical sales with predicted future demand trends.

The actual sales trend shows:

- strong demand during the initial months,
- a sharp decline during the mid-period,
- followed by gradual operational recovery.

The forecasted trend indicates:

future demand stabilization around
consistent sales levels

with relatively lower volatility compared to historical
fluctuations.

Possible Causes

Historical Demand Decline

The sharp decline in actual sales may be caused by:

- inventory shortages,
- supplier delays,
- operational disruptions,
- or reduced seasonal demand.

Forecast Stabilization

The forecast model predicts stable future demand due to:

- improving operational consistency,
- stabilized inventory availability,
- and recovery in pharmaceutical sales performance.

The prediction suggests:

reduced short-term demand volatility
in upcoming business periods.

Business Impact

Operational Impact

Accurate demand forecasting supports:

- better inventory planning,
- optimized warehouse allocation,
- improved supplier coordination,
- and efficient stock replenishment.

Financial Impact

Forecast-driven planning helps:

- reduce stock shortages,
- minimize overstocking,
- improve revenue predictability,
- and strengthen operational efficiency.

Reliable forecasting can significantly improve:

strategic decision-making and business
stability
across pharmaceutical operations.

Predictive Analysis / Future Outlook

The forecast indicates:

- stable future pharmaceutical demand,
- gradual business normalization,
- and improved operational continuity.

If current recovery trends continue:

future sales performance is expected to
remain consistent with moderate growth
potential

while avoiding severe historical fluctuations.

The forecasting model also suggests stronger predictability
in upcoming inventory and revenue planning cycles.

Recommendations

Forecast-Based Inventory Planning

Use forecasting outputs to optimize stock replenishment and warehouse allocation proactively.

Supplier Coordination

Align supplier procurement schedules with forecasted pharmaceutical demand patterns.

Continuous Forecast Monitoring

Update forecasting models regularly using new sales and operational data for improved prediction accuracy.

Risk Preparedness

Maintain safety stock for critical medicines to handle unexpected demand fluctuations.

Data-Driven Decision Making

Integrate forecasting insights into strategic planning for:

- inventory optimization,
- revenue management,
- and supply chain efficiency improvement.

Conclusion

This project successfully developed an end-to-end Pharmaceutical Analytics Framework using Python, SQL, and Power BI to analyze sales performance, inventory management, warehouse operations, supplier efficiency, and demand forecasting.

By applying Descriptive, Diagnostic, Predictive, and Prescriptive Analytics, the project transformed raw pharmaceutical data into actionable business insights that support data-driven decision-making, operational optimization, and supply chain efficiency.

The project demonstrates strong capabilities in:

- data analysis,
- statistical modeling,
- forecasting,
- dashboard development,
- and business intelligence